

# Lab 17: RNA-Seq Quantification

Omar Siddiqui (PID: A19093654)

2026-05-31

## RNA-Seq Quantification and PCA Analysis

First, we load the required libraries and import the abundance data using `tximport`:

```
setwd("/Users/omar")
library(tximport)

folders <- dir(pattern="SRR21568*")
files <- file.path(folders, "abundance.h5")
names(files) <- sub("_quant", "", folders)

txi <- tximport(files, type="kallisto", txOut=TRUE)
```

## Filtering

Remove transcripts with no reads and no variation:

```
to.keep <- rowSums(txi$counts) > 0
kset.nonzero <- txi$counts[to.keep,]

keep2 <- apply(kset.nonzero, 1, sd) > 0
x <- kset.nonzero[keep2,]
```

## PCA

```
pca <- prcomp(t(x), scale=TRUE)
summary(pca)
```

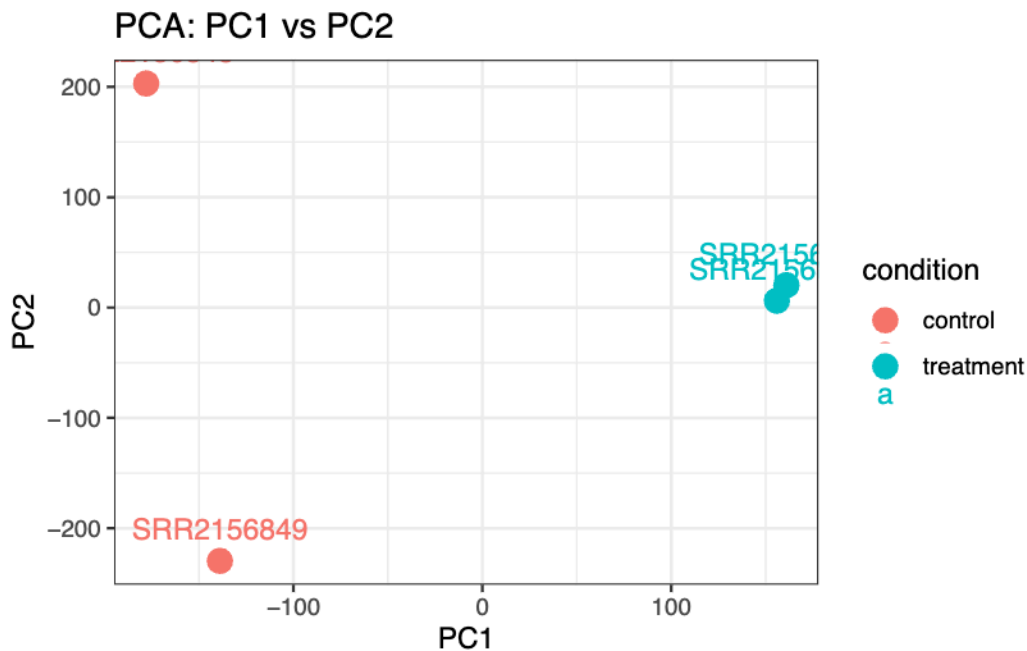
Importance of components:

|                        | PC1      | PC2      | PC3      | PC4   |
|------------------------|----------|----------|----------|-------|
| Standard deviation     | 183.6379 | 177.3605 | 171.3020 | 1e+00 |
| Proportion of Variance | 0.3568   | 0.3328   | 0.3104   | 1e-05 |
| Cumulative Proportion  | 0.3568   | 0.6895   | 1.0000   | 1e+00 |

```
library(ggplot2)

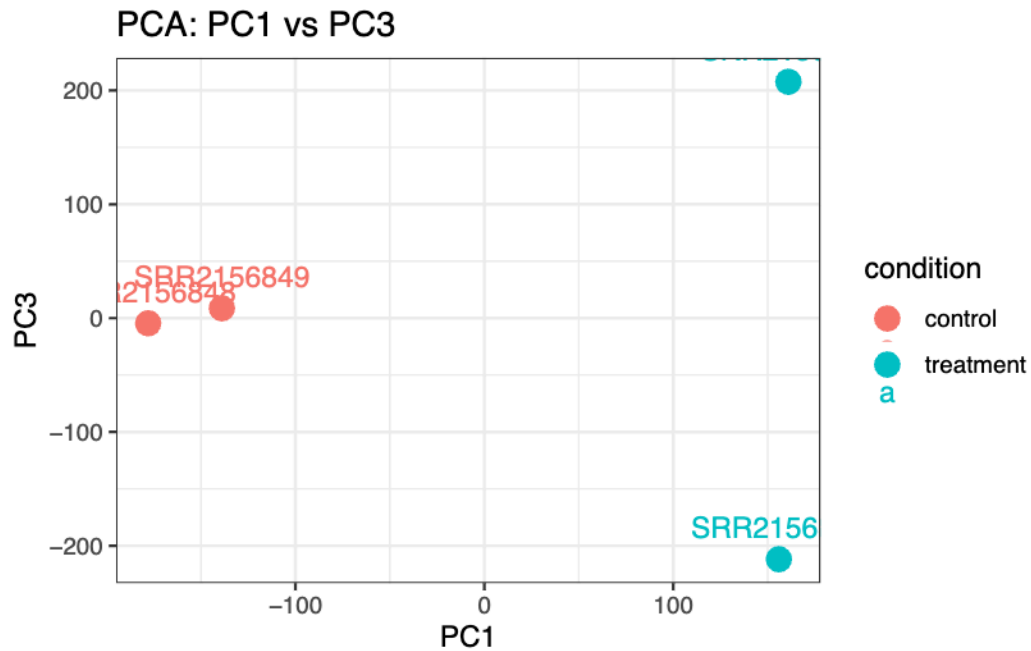
pca_df <- as.data.frame(pca$x)
pca_df$sample <- rownames(pca_df)
pca_df$condition <- c("control", "control", "treatment", "treatment")

# PC1 vs PC2
ggplot(pca_df, aes(x=PC1, y=PC2, color=condition, label=sample)) +
  geom_point(size=4) +
  geom_text(vjust=-1) +
  labs(title="PCA: PC1 vs PC2") +
  theme_bw()
```

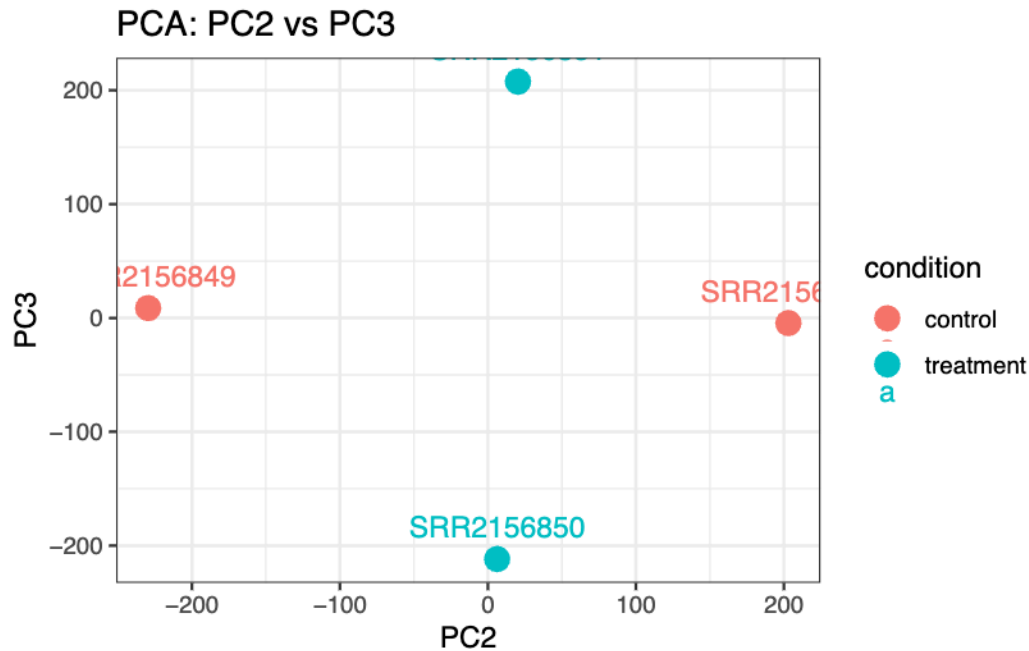


```
# PC1 vs PC3
ggplot(pca_df, aes(x=PC1, y=PC3, color=condition, label=sample)) +
  geom_point(size=4) +
```

```
geom_text(vjust=-1) +
labs(title="PCA: PC1 vs PC3") +
theme_bw()
```



```
# PC2 vs PC3
ggplot(pca_df, aes(x=PC2, y=PC3, color=condition, label=sample)) +
  geom_point(size=4) +
  geom_text(vjust=-1) +
  labs(title="PCA: PC2 vs PC3") +
  theme_bw()
```



## Lab Questions

### Q: What do the abundance.tsv files contain?

Each file contains columns for transcript ID, length, effective length, estimated counts, and TPM. Most transcripts have zero counts — only a subset are expressed in each sample.

### Q: PCA Interpretation

PC1 separates the two control samples (blue) from the two CRISPR treatment samples (red), indicating a strong difference in transcript abundance between conditions. PC2 separates the two control replicates from each other, and PC3 separates the two treatment replicates.